Hedging Bitcoin with Commodity Futures: An Analysis with Copper, Gas, Gold, and Crude Oil Futures*

Young C. Joo,† Sung Y. Park‡

Abstract

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Keywords: DCC-GARCH, ADCC-GARCH, Hedging, Commodity Futures, Cryptocurrency

JEL classification: G11, G17, C32, C58

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1 Introduction

In recent years, cryptocurrency has experienced rapid development and received considerable attention. It is considered an alternative investment vehicle, providing a highly profitable opportunity and great flexibility. These apparent characteristics of cryptocurrency have promoted investors to construct new investment strategies that differ from traditional investments (Pal and Mitra, 2009; Ma, Tian, Hsiao, and Deng 2022; Bergsli, Lind, Molnar, and Polasik, 2022). However, cryptocurrency has higher volatility and risk than traditional investment assets and, therefore, requires more attention from investors. The increasing interest in cryptocurrency has led to a need for research on the hedging strategy that could offset or mitigate risk with traditional investment.

Investment strategies that include cryptocurrency are widely used to achieve high returns. Although cryptocurrencies are highly volatile and risky, given their previously mentioned characteristics, the return on portfolios that include cryptocurrencies is relatively high (Gandal and Halaburda, 2014; Eisl, Gasser, and Weinmayer, 2015). Several studies have investigated portfolios that include virtual currencies and also hedging assets with cryptocurrencies (Bouri, Azzi, and Haubo, 2016; Baur, Lee, and Hong, 2018; Park, 2022). For instance, Dyhberg (2015a, b) studies the hedging capability of cryptocurrency with commodities, equities, and the US dollar and finds that Bitcoin acts as a hedge against uncertainty like other equities. Recent research shows that due to the weak positive or even negative correlation between cryptocurrency and equities, cryptocurrency acts as an effective diversifier and hedge in portfolios (Bouri, Molnar, Azzi, Roubaud, and Hagfors, 2017).

An asset can be defined as a hedge, diversifier, and haven according to their relationships (Baur and Lucey, 2010; Ratner and Chiu, 2013). If an asset is uncorrelated or negatively correlated with another asset or portfolio, it is a hedge. However, if the asset shows this relationship in only the stress or turmoil of market conditions, it is a haven. Unlike a hedge and haven, a diversifier is an asset that, on average, positively correlates with another asset or portfolio. Gold has often been

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used as a hedge for risky assets, and many studies have found that it is a hedge and haven for stocks and equities (Baur and Lucey, 2010; Baur and McDermott, 2010; Bredin, Conlon, and Poti, 2015; Lucey and Li, 2014; Ciner, Gurdgiev, and Lucey, 2013). However, recently, some investors have interest in alternative assets other than traditional investment assets, so cryptocurrencies are considered an alternative investment asset. Given the surge in interest, cryptocurrencies have been examined to identify their role and hedging capabilities against assets (Dyhrberg, 2016; Bouri, Gupta, Tiwari, and Roubaud, 2017; Bouri, Molnár, Azzi, Roubaud, and Hagfors, 2017; Corbet, Meegan, Larkin, Lucey, and Yarovaya, 2018).

This study aims to investigate the relationship between cryptocurrency and commodity futures and determine whether cryptocurrency can be classified as a hedge against commodity futures. We attempt to capture the dynamic conditional correlations among assets by employing dynamic conditional correlation (DCC) and asymmetric dynamic conditional correlation (ADCC) multivariate generalized autoregressive conditional heteroscedasticity (MGARCH) models. These models help uncover behavior of optimal hedge ratios and hedging effectiveness using conditional volatility and correlation estimates.

In analyzing the relationships between commodity futures and Bitcoin, it is of great interest examining the hedging capacities of traditional futures investment against cryptocurrency. To this end, we consider the four most commonly invested futures in financial markets: copper, gas, gold, and crude oil. To our knowledge, no other study has investigated the hedging capacities between cryptocurrency and commodity futures. We construct one-step-ahead hedge ratios using a rolling window scheme based on dynamic conditional correlation models, enabling us to analyze the effect of changing asset variability on the hedge ratios. Moreover, we evaluate the performance of the hedged portfolios according to the amount of variance reduction compared with unhedged portfolios.

Our study produced several important results. First, we find that returns of the futures regard-

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ing gas and gold are negatively correlated with Bitcoin, which suggest that these two futures have hedge benefits. Furthermore, from the results of the one-step-ahead optimal hedge ratios, we find that all hedge ratios dropped in 2018 due to economic stagnation; however, there was a large increase in early 2020 due to the shocks induced by COVID-19. These results indicate that hedge ratios need to be regularly updated. Finally, the results of hedge effectiveness show that, in the case of short-term forecast periods, gold provides the most effective hedge for Bitcoin. In the long term, copper shows higher hedge effectiveness than other commodities. However, crude oil has a poor hedge performance due to its positive correlations.

Our study makes several contributions to the literature. First, we estimate time-varying optimal hedge ratios between Bitcoin and four well-known commodity futures using DCC-type multivariate GARCH models. Second, we construct one-step-ahead optimal hedge ratios using a rolling window method to capture a more comprehensive out-of-sample behavior of optimal hedge ratios. Third, we conduct an extensive analysis to explore hedge performance by comparing the hedge effectiveness for various numbers of model refit (the number of periods to re-estimate the model) and lengths of the forecast horizon (the number of one-step-ahead forecasts). This analysis provides some results on how robust are the hedging effectiveness to change in numbers of model refit and lengths forecast horizon. The results of our analysis provide useful insights to investors, practitioners, and researchers.

The remainder of our study is organized as follows: Section 2 describes the DCC- and ADCC-MGARCH models and introduces the optimal hedge ratios. Next, Section 3 describes the data and presents the estimation results. Finally, Section 4 summarizes and concludes.

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2 The Models

2.1 DCC- and ADCC-MGARCH Models

We consider the dynamic conditional correlation (DCC, Engle, 2002) and asymmetric dynamic conditional correlation (ADCC, Cappiello, Engle, and Sheppard, 2006) MGARCH models to capture the volatility dynamics, conditional correlations, and hedge ratios of Bitcoin and copper, gas, gold, and crude oil. From the DCC-MGARCH model, we can capture dynamic conditional correlation and hedge ratios between assets and four commodity futures. The ADCC-MGARCH model helps capture a change in conditional correlation and hedge ratios during the market downturn.

Let $\boldsymbol{r}_t$ be a $n \times 1$ vector of asset returns (i.e., returns of Bitcoin and commodity futures). The ARMA(1, 1) process for $\boldsymbol{r}_t$ conditional on the information set Ω_{t-1} can be written as follows:

$$\boldsymbol{r}_t = \mu + a\boldsymbol{r}_{t-1} + b\boldsymbol{\varepsilon}_{t-1} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t|\Omega_{t-1} \sim N(0, H_t). \tag{2.1}$$

The errors can be expressed as follows:

$$\boldsymbol{\varepsilon}_t = H_t^{1/2}\boldsymbol{u}_t, \tag{2.2}$$

where H_t is the conditional covariance matrix of $\boldsymbol{r}_t$ and $\boldsymbol{u}_t$ is a $n \times 1$ *i.i.d.* random vector of errors.

The DCC-MGARCH model (Engle, 2002) is defined as follows:

$$\boldsymbol{\varepsilon}_t|\Omega_{t-1} \sim N(0, H_t), \quad t = 1, 2, \cdots, T, \tag{2.3}$$

$$H_t = D_t R_t D_t, \tag{2.4}$$

where H_t is a $n \times n$ conditional covariance matrix of $\boldsymbol{r}_t$, $R_t = \{diag(Q_t)^{-1/2}\}Q_t\{diag(Q_t)^{-1/2}\}$ is the conditional correlation matrix, and $D_t = diag\left(h_{1t}^{1/2}, \cdots, h_{nt}^{1/2}\right)$ is a diagonal matrix with time-varying conditional standard deviation. Since the conditional variance h_{it} follows univariate GARCH models, it can be written as follows:

$$h_{it} = \omega_i + \sum_{k=1}^{p} \alpha_{ik}\varepsilon_{it-k}^2 + \sum_{l=1}^{q} \beta_{il}h_{it-l}. \tag{2.5}$$

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A symmetric positive definite matrix Q_t is given by:

$$Q_t = (1 - \xi_1 - \xi_2)\bar{Q} + \xi_1 u_{t-1} u'_{t-1} + \xi_2 Q_{t-1}, \tag{2.6}$$

where $\bar{Q}$ is the $n \times n$ unconditional correlation matrix of the standardized residuals $u_{it} = \varepsilon_{it}/h_{it}^{1/2}$ and $u_t = (u_{1t}, \cdots, u_{nt})$. The parameters ξ_1 and ξ_2 are non-negative scalar parameters satisfying $\xi_1 + \xi_2 < 1$, which guarantees that $Q_t > 0$. These parameters are used to capture the effects of previous shocks on the current dynamic conditional correlation. If $\xi_1 = \xi_2 = 0$, then Q_t is equal to the constant conditional covariance matrix.

For the ADCC-MGARCH model (Cappiello, Engle, and Sheppard, 2006), the individual GARCH model has an asymmetric term:

$$h_{it} = \omega_i + \sum_{k=1}^{p} \alpha_{ik} \varepsilon_{it-k}^2 + \sum_{l=1}^{q} \beta_{il} h_{it-l} + \sum_{j=1}^{m} d_{ij} \varepsilon_{it-j}^2 I(\varepsilon_{it-j}), \tag{2.7}$$

where $I(\varepsilon_{it-j})$ denotes an indicator function that is equal to one if $\varepsilon_{it-j} < 0$ and zero otherwise. For this specification, a positive d_i implies that previous negative residuals tend to increase the variance more than positive ones, indicating the leverage effect.

For the ADCC-MGARCH model, the dynamics of Q_t are given by:

$$Q_t = (1 - \xi_1 - \xi_2)\bar{Q} - \xi_3 \bar{M} + \xi_1 u_{t-1} u'_{t-1} + \xi_2 Q_{t-1} + \xi_3 \zeta_{t-1} \zeta'_{t-1}, \tag{2.8}$$

where $\zeta_t = I[u_t < 0] \otimes u_t$ are zero-threshold standardized errors which are equal to u_t when less than zero and zero otherwise and $\bar{M} = E[\zeta_t \zeta'_t]$. $I[\cdot]$ is the indicating function and $\otimes$ denotes the element-by-element multiplication operator. The ADCC-MGARCH model also has a sufficient and necessary condition, $\xi_1 + \xi_2 + \delta \xi_3 < 1$, where δ denotes the maximum eigenvalue, $\bar{Q}^{-1/2} \bar{M} \bar{Q}^{-1/2}$. The parameter ξ_3 is the asymmetric term indicating market downturn periods.

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2.2 Optimal Conditional Hedge Ratio

One of the main purposes for futures market participants is to hedge the risk to achieve their individual goals. Specifically, in the case of a risky asset, we usually want to protect exposure to asset price fluctuations. The return on a portfolio of a spot and futures position can be represented as follows:

$$R_{Ht} = R_{St} - \gamma_t R_{Ft}, \tag{2.9}$$

where R_{Ht} is the return on the hedged portfolio to the Bitcoin, R_{St} represents returns on the spot position (return of Bitcoin), R_{Ft} denotes the returns on the futures position (returns of the futures), and γ_t is the time-varying hedge ratio indicating the number of futures contracts. Specifically, if the investor takes long in the spot position then the hedge ratio indicates the number of futures contracts that must be sold for each unit of spot commodity on which price risk is borne. The variance of the hedged portfolio conditional on the information set available at time $t-1$ is given by:

$$Var(R_{Ht}|\Omega_{t-1}) = Var(R_{St}|\Omega_{t-1}) - 2\gamma_t Cov(R_{Ft}R_{St}|\Omega_{t-1}) + \gamma_t^2 Var(R_{Ft}|\Omega_{t-1}), \tag{2.10}$$

where $Var(R_{St}|\Omega_{t-1})$, $Var(R_{Ft}|\Omega_{t-1})$, and $Cov(R_{Ft}R_{St}|\Omega_{t-1})$ are the conditional variance and covariance of spot and futures returns, respectively. The optimal hedge ratios can be obtained by minimizing the conditional variance (risk) of the hedged portfolio returns, i.e., $\min_{\gamma_t}[Var(R_{Ht}|\Omega_{t-1})]$. The optimal conditional hedge ratio can be obtained by taking the partial derivative of the variance with respect to γ_t and setting the expression equal to zero (Baillie and Myers, 1991) as follows:

$$\gamma_t|\Omega_{t-1} = \frac{Cov(R_{St}, R_{Ft}|\Omega_{t-1})}{Var(R_{Ft}|\Omega_{t-1})}. \tag{2.11}$$

In order to estimate the time-varying optimal hedge ratio conditional variance and covariance need to be estimated. The conditional covariance matrix estimates from the multivariate conditional volatility model of GARCH models can be used to construct hedge ratios (Kroner and Sul-

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tan, 1993). The optimal conditional hedge ratio between spot and futures positions is as follows:

$$\hat{\gamma}_t|\Omega_{t-1} = \frac{\hat{h}_{SFt}}{\hat{h}_{Ft}}, \tag{2.12}$$

where $\hat{h}_{SFt}$ is the estimated conditional covariance between spot and futures positions and $\hat{h}_{Ft}$ is the estimated conditional variance of futures returns. Some studies propose flexible hedging models to achieve better hedging performance in terms of hedging effectiveness (Chang, McAller, and Tansuchat, 2011; Park and Jei, 2010; Ku, Chen, and Chen, 2007).

We evaluate the performance of the hedged portfolio obtained from different MGARCH models using the hedging effectiveness (HE) index by the amount of the variance reduction for any hedged portfolio compared with the unhedged portfolio.

$$HE = \frac{Var(R_{Ut}) - Var(R_{Ht})}{Var(R_{Ut})}, \tag{2.13}$$

where $Var(R_{Ht})$ is the variance of the return of hedged portfolio, R_{Ht}, and $Var(R_{Ut})$ is the variance of the return of unhedged portfolio, R_{Ut}. A higher value of the HE index implies larger risk reduction, indicating higher hedging effectiveness and an outperformed hedging strategy.

Out-of-sample hedge ratios are constructed using a rolling window analysis. At time period t, a one-period-ahead conditional volatility forecast is made and these forecasts are used to construct a one-step-ahead hedge ratio. These forecasted hedge ratios are then used to constructing the hedged portfolio. We used 1848 estimation window observations to construct 1000 one-step-ahead hedge ratios. MGARCH models are refit every 20 observations.

3 Empirical Analysis

3.1 Data

Our dataset, obtained from Datastream, comprises the daily price of Bitcoin (BTC) and futures of copper (Copper), natural gas (Gas), gold (Gold), and Brent crude (Oil) for the period August 1,

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2010, to November 18, 2021. It yields a total of 2848 observations.

[Figure 1]

We compute the return series (first difference of the natural logarithmic series), $\ln(p_t/p_{t-1})$, where p_t and p_{t-1} are the prices at time t and $t-1$, respectively. Figure 1 plots the price and return series to show changes in prices and returns over time. Prices of copper and gold display similar movements, and gas and oil also share similar trends. In addition, the return series of copper and gold show volatility clustering around 2012-2013. The return series of gas and oil also display volatility clustering around 2020.

[Table 1]

Table 1 provides the descriptive statistics of the return series. To assess the stationary properties of the series, we apply two unit root tests on the return series: the Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) tests. The ADF test results confirm that all return series reject the null hypothesis that the series has a unit root. The PP test results show that all return series are stationary, thus rejecting the null hypothesis. Therefore, we use the return series for the empirical analysis.

All return series have positive mean values. Specifically, BTC has the highest mean value of price return at 0.465 followed by Gold, Copper, Gas, and Oil. Although BTC and Gold have negative skewness (Skew), Copper, Gas, and Oil have positive skewness. Kurtosis (Kurt) of return series are greater than 3 except for copper. Thus most returns are skewed and leptokurtic. The test results of Jarque–Bera (JB) for all return series are statistically significant, indicating that the normality assumption for all return series is rejected. These test statistics indicate that all the return series exhibit a high degree of non-normality. We consider these return series with non-normal distribution by applying univariate and multivariate student t-distributions for GARCH model es-

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timations to generate more robust estimation results. Table 2 presents unconditional correlations of variables. It shows that BTC return has significantly positive correlations with Copper returns. These are the strongest correlations across all the asset returns. In contrast, BTC return correlates negatively and significantly with Gas return.

[Table 2]

3.2 Estimation Results

In the following analysis, we analyze returns of BTC, Copper, Gas, Gold, and Oil using DCC- and ADCC-MGARCH models to identify the time-varying relationship and the hedge effects between assets. The DCC- and ADCC-MGARCH models can assist in understanding the dynamics of conditional correlation between assets and be useful in explaining the flow of the optimal hedge ratios over time. The objectives of this analysis are the construction of optimal hedge ratios between the high-risk asset and well-known futures and the comparison of the hedge effects between the assets. Being well-known, DCC- and ADCC-MGARCH models are estimated by taking two-step estimation procedures. We set the specifications of mean and variance equations for the estimation. Based on some well-known information criteria (AIC, BIC, SIC, and HQ), the ARMA (1,1) for the conditional mean equation is selected.[1] Based on these specifications, we report the results of the DCC- and ADCC- MGARCH models with a multivariate t-distribution.

[Table 3]

In Table 3, we report the estimation results of the DCC- and ADCC-MGARCH models. Firstly, we observe that the estimated coefficients, $\hat{\alpha}$ and $\hat{\beta}$ in the conditional variance equation indicates the short- and long-term persistence and evidence of volatility clustering. In all cases, long-term

[1] All other results are available in the supplementary materials.

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persistence ($\hat{\beta}$) is higher than short-term persistence ($\hat{\alpha}$), and they are statistically significant. In addition, it can be seen that there exists a statistically significant positive asymmetric effect, $\hat{\gamma}$, for Copper and Oil, indicating that negative shocks tend to increase the variance of the returns more than positive shocks. The coefficient on the shape parameter, $\hat{\nu}$, indicates the degree of freedom of the multivariate t-distribution. Thus, as the value of $\hat{\nu}$ approaches infinity, the shape of the distributions for the return series approaches that of a normal distribution. We find that each return series, especially Bitcoin, has a low value for the estimated shape parameter, indicating that the distribution of the return series has a heavy tail. The $\hat{\xi}_1$ and $\hat{\xi}_2$ are positive and statistically significant. The sum of $\hat{\xi}_1$ and $\hat{\xi}_2$ is less than one indicating dynamic conditional correlation is mean reverting.

[Figure 2]

Figure 2 shows the one-step-ahead dynamic conditional correlations for assets from the M GARCH models. The results show that the conditional correlations estimated from the ADCC-MGARCH model show a higher level than those from the DCC-MGARCH model for all cases. Among the four assets, the lowest conditional correlation occurs for BTC/Gas, and, on the other hand, the highest conditional correlation occurs for BTC/Gold. We also find that gold and gas are mostly negatively correlated with Bitcoin, implying that these two assets may provide hedging opportunities. This result indicates that gold and gas have hedge benefits against Bitcoin. We also find that BTC/Copper, BTC/Gold, and BTC/Oil show a sharp decline in dynamic conditional correlation in early 2020, reflecting the increase in copper, gold, and oil prices and the decline in Bitcoin price.

[Figure 3]

Figure 3 plots the optimal conditional hedge ratio estimates for Bitcoin, copper, gas, gold, and

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oil calculated from DCC- and ADCC-GARCH models. Note that the hedge ratios from the the estimated DCC- GARCH model are similar to those of the ADCC-GARCH model, but the DCC hedge ratios are generally lower than ADCC hedge ratios. All hedge ratios dropped in early 2018 due to economic stagnation and stock market issues, like global trade spats, rising interest rates, and Brexit uncertainty. In addition, the hedge ratios for BTC/Copper, BTC/Gas, and BTC/Gold experienced a large increase in early 2020 due to the impact of COVID-19 on global financial markets.

We evaluate hedging effectiveness under various cases by calculating the hedge ratios and variance reduction measure under different forecast horizons and numbers of model refit (the number of periods to re-estimate the model). Table 4 presents additional hedge ratios and effectiveness results from models refit every 10, 20, and 50 days. From the left side of the Table 4, the hedge effectiveness values are presented under the number of model refit periods, 10, 20, and 50 days. For each number of model refit, we also consider five different forecast lengths, namely 100, 500, 1,000, 1,500, and 2,000. Furthermore, we report the mean, minimum, and maximum of the estimated hedge ratios and hedge effectiveness, HE, from the DCC- and ADCC-MGARCH models.

[Table 4]

For a forecast length of 100 days, BTC/Gold shows the highest HE values for all model refit levels. This result indicates that gold provides the most effective hedge for Bitcoin in the short term. For BTC/Copper and BTC/Gas, HE values are mostly positive. In the short-term forecast lengths, the copper shows higher HE values. However, in the long-term cases, gas has higher HE values. Unlike other assets, BTC/Oil shows a negative HE value for all cases, indicating that it performs worse than an unhedged portfolio. This result could occur because they are positively correlated. For all assets, HE values for DCC are higher than ADCC, which shows that DCC is better suited to capture the volatile nature of asset prices.

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4 Concluding Remarks

In this paper, we used the multivariate GARCH models to explore the usefulness of using copper, gas, gold, and oil futures to hedge the risk associated with cryptocurrency investing. Using the DCC- and ADCC-MGARCH models, we evaluate the dynamic conditional correlations and one-step-ahead volatility to calculate and compare optimal hedge ratios and hedge effectiveness.

In analyzing the relationships between commodity futures and Bitcoin, examining the hedging capacities of traditional futures investment for cryptocurrency is of great interest. To this end, we consider four of the most commonly invested futures in financial markets: copper, gas, gold, and crude oil. To our knowledge, no other study has investigated the out-of-sample hedging performance between cryptocurrency and commodity futures. We construct one-step-ahead hedge ratios using a rolling window scheme based on dynamic conditional correlation models, enabling us to analyze the changing variability in Bitcoin and commodity futures. Moreover, we evaluate the performance of the hedged portfolio by the amount of variance reduction compared with the unhedged portfolio.

From the empirical analysis, we find that futures of gas and gold have hedge benefits. However, crude oil shows poor hedging performance. We also find that dynamic conditional correlations between Bitcoin and copper, gold, and oil decline sharply in early 2020, reflecting the increase in copper, gold, and oil prices and the decline in Bitcoin price. From the results of one-step-ahead hedge ratios, we find that hedge ratios decreased in 2018 due to economic stagnation but increased in early 2020 due to the impact of COVID-19 shocks, indicating that hedge ratios need to be updated regularly.

The results of hedge effectiveness show that, for each hedge, DCC produces better hedging

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effectiveness. This result indicates that DCC is better suited to capture the volatile nature of futures prices. In the case of short-term forecast periods, gold provides the most effective hedge for Bitcoin, while copper provides better hedging effectiveness in the long term. However, oil has poor hedge performance due to its positive correlations.

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This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4355765

References

Baillie, R. T., and R. Myers (1991): "Bivariate GARCH Estimation of the Optimal Commodity Futures Hedge," *Journal of Applied Econometrics*, 6, 109–124.

Baur, D., and T. McDermott (2010): "Is Gold a Safe Haven? International Evidence," *Journal of Banking and Finance*, 34, 1886–1898.

Baur, D. G., K. Hong, and A. D. Lee (2018): "Bitcoin: Medium of Exchange or Speculative Assets?," *Journal of International Financial Markets, Institutions and Money*, 54, 177–189.

Bergsli, L. O., A. F. Lind, P. Molnar, and M. Polasik (2022): "Forecasting Volatility of Bitcoin," *Research in International Business and Finance*, 29, 101540.

Bour, D. G., and B. M. Lucey (2010): "Is Gold a Hedge or a Safe Haven? An Analysis of Stocks, Bonds and Gold," *Financial Review*, 45, 217–229.

Bouri, E., G. Azzi, and A. H. Dyhrberg (2016): "On the return-volatility relationship in the Bitcoin market around the price crash of 2013," Economics Discussion Papers 2016-41, 1–19.

Bouri, E., R. Gupta, A. Tiwari, and D. Roubaud (2017): "Does Bitcoin Hedge Global Uncertainty? Evidence from Wavelet-based Quantile-in-Quantile Regressions," *Finance Research Letters*, 23, 87–95.

Bouri, E., P. Molnar, G. Azzi, D. Roubaud, and L. I. Hagfors (2017): "On the Hedge and Safe Haven Properties of Bitcoin: Is it Really More Than a Diversifier?," *Finance Research Letters*, 20, 192–198.

Bredin, D., T. Conlon, and V. Poti (2015): "Does Gold Glitter in the Long-run? Gold as a Hedge and Safe Haven Across Time and Investment Horizon," *International Review of Financial Analysis*, 41, 320–328.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4355765

Cappiello, L., R. Engle, and K. Sheppard (2006): "Asymmetric Dynamics in the Correlations of Global Equity and Bond Returns," *Journal of Financial Econometrics*, 4, 537–572.

Chang, C.-L., M. McAleer, and R. Tansuchat (2011): "Crude Oil Hedging Strategies using Dynamic Multivariate GARCH," *Energy Economics*, 33, 912–923.

Ciner, C., C. Gurdgiev, and B. M. Lucey (2013): "Hedges and Safe Havens: An Examination of Stocks, Bonds, Gold, Oil and Exchange Rates," *International Review of Financial Analysis*, 29, 202–211.

Corbet, S., A. Meegan, C. Larkin, B. Lucey, and L. Yarovaya (2018): "Exploring the Dynamic Relationships between Cryptocurrencies and Other Financial Assets," *Economics Letters*, 165, 28–34.

Dyhrberg, A. H. (2015a): "Bitcoin, Gold and the Dollar - A GARCH Volatility Analysis," *Finance Research Letters*, 16, 85–92.

——— (2015b): "Hedging Capabilities of Bitcoin. Is it the Virtual Gold?," *Finance Research Letters*, 16, 139–144.

Dyhrberg, A. H. (2016): "Bitcoin, Gold and the Dollar - A GARCH Volatility Analysis," *Finance Research Letters*, 16, 85–92.

Eisl, A., S. M. Gasser, and K. Weinmayer (2015): "Caveat Emptor: Does Bitcoin Improve Portfolio Diversification?," SSRN Working Paper, 1–21.

Engle, R. (2002): "Dynamic Conditional Correlation: A Simple Class of Multivariate Generalized Autoregressive Conditional Heteroskedasticity Models," *Journal of Business and Economic Statistics*, 20, 339–350.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4355765

Gandal, N., and H. Halaburda (2014): "Competition in the Cryptocurrency Market," CEPR Discussion Paper No. DP10157, 1–11.

Kroner, K. F., and J. Sultan (1993): "Time-Varying Distributions and Dynamic Hedging with Foreign Currency Futures," *Journal of Financial and Quantitative Analysis*, 28, 535–551.

Ku, Y.-H. H., H.-C. Chen, and K.-H. Chen (2017): "On the Application of the Dynamic Conditional Correlation Model in Estimating Optimal Time-varying Hedge Ratios," *Applied Economics Letters*, 14, 503–509.

Lucey, B. M., and S. Li (2015): "What Precious Metals Act as Safe Havens, and When? Some US Evidence," *Applied Economics Letters*, 22, 35–45.

Ma, C., Y. Tian, S. Hsiao, and L. Deng (2022): "Monetary policy shocks and Bitcoin prices," *Research in International Business and Finance*, 62, 101711.

Pal, D., and S. K. Mitra (2019): "Hedging Bitcoin with Other Financial Assets," *Finance Research Letters*, 30, 30–36.

Park, B. J. (2022): "The COVID-19 Pandemic, Volatility, and Trading Behavior in the Bitcoin Futures Market," *Research in International Business and Finance*, 59, 101519.

Park, S. Y., and S. Y. Jei (2010): "Estimation and Hedging Effectiveness of Time-varying Hedge Ratio: Flexible Bivariate GARCH Approaches," *Journal of Futures Markets*, 30, 71–99.

Ratner, M., and C. Chiu (2013): "Hedging Stock Sector Risk with Credit Default Swaps," *International Review of Financial Analysis*, 30, 18–25.

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Table 1: Descriptive Statistics for Return Series

	Mean	Median	S.D.	Skew	Kurt	ADF	PP	JB
BTC	0.465	0.000	7.470	2.447	69.870	-12.974***	-56.315***	581940.036***
Copper	0.008	0.000	1.368	-0.078	2.210	-14.022***	-55.642***	582.055***
Gas	0.001	0.000	2.983	0.259	3.852	-15.111***	-55.888***	1791.971***
Gold	0.016	0.026	1.022	-0.654	6.949	-13.798***	-54.872***	5932.182***
Oil	0.000	0.009	2.285	-0.554	26.201	-13.379***	-53.387***	81582.884***

Notes: The numbers in parentheses are standard errors. *, **, and *** denote significance at 10%, 5%, and 1%, respectively.

Table 2: Correlation Matrix between Return Series

	BTC	Copper	Gas	Gold	Oil
BTC	1.000	0.040***	-0.030*	0.010	0.040***
Copper	0.040***	1.000	0.050***	0.260***	0.280***
Gas	-0.030*	0.050***	1.000	0.010	0.100***
Gold	0.010	0.260***	0.010	1.000	0.100***
Oil	0.040***	0.280***	0.100***	0.100***	1.000

Notes: The numbers in parentheses are standard errors. *, **, and *** denote significance at 10%, 5%, and 1%, respectively.

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Table 3: Estimation Results of DCC and ADCC Models

Variables	DCC		ADCC	
	Coef.	S.E.	Coef.	S.E.
μ_{BTC}	0.239***	(0.071)	0.240***	(0.077)
a_{BTC}	-0.406	(0.257)	-0.400*	(0.233)
b_{BTC}	0.398	(0.258)	0.391	(0.235)
ω_{BTC}	0.981	(2.482)	0.952	(2.904)
α_{BTC}	0.223***	(0.056)	0.229***	(0.042)
β_{BTC}	0.776***	(0.178)	0.778***	(0.211)
γ_{BTC}			-0.017	(0.067)
λ_{BTC}	2.897***	(0.126)	2.900***	(0.144)
μ_{Copper}	0.010	(0.021)	0.004	(0.021)
a_{Copper}	-0.192	(0.221)	-0.202	(0.208)
b_{Copper}	0.132	(0.224)	0.143	(0.211)
ω_{Copper}	0.014***	(0.005)	0.014***	(0.005)
α_{Copper}	0.030***	(0.003)	0.016***	(0.005)
β_{Copper}	0.963***	(0.001)	0.965***	(0.000)
γ_{Copper}			0.025***	(0.009)
λ_{Copper}	5.604***	(0.557)	5.710***	(0.571)
μ_{Gas}	-0.020	(0.037)	-0.014	(0.042)
a_{Gas}	0.887***	(0.159)	0.482	(0.392)
b_{Gas}	-0.905***	(0.148)	-0.520	(0.383)
ω_{Gas}	0.156***	(0.039)	0.152***	(0.039)
α_{Gas}	0.073***	(0.009)	0.081***	(0.011)
β_{Gas}	0.911***	(0.010)	0.914***	(0.010)
γ_{Gas}			-0.022	(0.018)
λ_{Gas}	6.809***	(0.771)	6.736***	(0.755)
μ_{Gold}	0.035*	(0.014)	0.036***	(0.014)
a_{Gold}	-0.098	(0.191)	-0.099	(0.189)
b_{Gold}	0.049	(0.191)	0.048	(0.189)
ω_{Gold}	0.008***	(0.002)	0.008***	(0.002)
α_{Gold}	0.029***	(0.003)	0.039***	(0.008)
β_{Gold}	0.966***	(0.001)	0.965***	(0.001)
γ_{Gold}			-0.015	(0.010)
λ_{Gold}	3.628***	(0.258)	3.625***	(0.258)
μ_{Oil}	0.018	(0.026)	0.018	(0.026)
a_{Oil}	0.183	(0.156)	0.180	(0.153)
b_{Oil}	-0.218	(0.154)	-0.216	(0.151)
ω_{Oil}	0.051***	(0.019)	0.047***	(0.017)
α_{Oil}	0.082***	(0.014)	0.033***	(0.012)
β_{Oil}	0.913***	(0.014)	0.922***	(0.015)
γ_{Oil}			0.074***	(0.018)
λ_{Oil}	4.110***	(0.330)	4.316***	(0.364)
ξ_1	0.008*	(0.004)	0.008**	(0.004)
ξ_2	0.979***	(0.017)	0.979***	(0.017)
ξ_3			0.001	(0.001)
ν	5.677***	(0.201)	5.759***	(0.205)
AIC	20.563		20.555	
BIC	20.663		20.668	
SIC	20.562		20.554	
H-Q	20.599		20.596	
Loglik	-29223		-29206	

Notes: The numbers in parentheses are standard errors. *, **, and *** denote significance at 10%, 5%, and 1%, respectively.

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Table 4: Hedge Ratios and Hedging Effectiveness

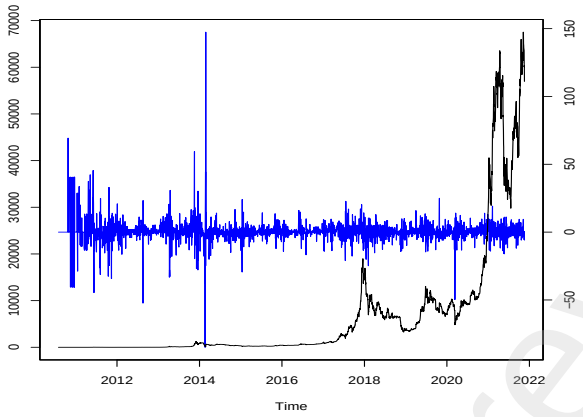
Model	Variable	re=10					re=20					re=50				
		fl=100	fl=500	fl=1000	fl=1500	fl=2000	fl=100	fl=500	fl=1000	fl=1500	fl=2000	fl=100	fl=500	fl=1000	fl=1500	fl=2000
		BTC/COPPER														
DCC	Mean	0.188	0.236	0.156	0.123	0.103	0.184	0.235	0.154	0.121	0.107	0.169	0.242	0.151	0.116	0.111
	Min	0.048	-0.166	-0.361	-0.908	-20.648	0.039	-0.166	-0.361	-0.908	-20.027	0.039	-0.176	-0.446	-0.908	-20.648
	Max	0.367	1.900	1.335	1.336	3.282	0.367	1.878	1.284	1.261	4.607	0.332	2.428	1.341	1.220	5.570
	HE	0.004	-0.004	0.001	-0.001	0.003	0.004	-0.003	0.001	-0.001	0.007	0.003	-0.010	0.001	-0.001	0.007
ADCC	Mean	0.196	0.249	0.173	0.130	0.115	0.192	0.250	0.173	0.129	0.117	0.178	0.252	0.171	0.128	0.120
	Min	0.056	-0.155	-0.365	-0.881	-22.794	0.048	-0.155	-0.365	-0.881	-23.015	0.048	-0.164	-0.462	-0.881	-22.794
	Max	0.370	1.890	1.480	1.517	2.658	0.370	1.908	1.421	1.458	3.776	0.336	1.996	1.452	1.411	4.504
	HE	0.004	-0.004	0.000	-0.002	0.000	0.004	-0.004	0.000	-0.002	0.004	0.003	-0.005	0.000	-0.001	0.005
		BTC/GAS														
DCC	Mean	-0.055	-0.048	-0.039	-0.048	-0.043	-0.054	-0.048	-0.039	-0.047	-0.043	-0.054	-0.056	-0.037	-0.046	-0.043
	Min	-0.154	-0.264	-0.398	-0.664	-2.432	-0.154	-0.266	-0.398	-0.664	-2.582	-0.146	-0.617	-0.398	-0.664	-2.432
	Max	0.003	0.118	0.597	0.378	0.434	0.003	0.118	0.597	0.378	0.434	0.009	0.118	0.676	0.394	0.459
	HE	0.002	0.002	-0.004	-0.001	0.009	0.002	0.003	-0.004	-0.001	0.009	0.002	0.004	-0.004	-0.001	0.003
ADCC	Mean	-0.050	-0.041	-0.031	-0.043	-0.032	-0.049	-0.042	-0.031	-0.043	-0.033	-0.048	-0.045	-0.031	-0.042	-0.034
	Min	-0.140	-0.224	-0.381	-0.709	-2.594	-0.140	-0.223	-0.381	-0.709	-2.873	-0.129	-0.320	-0.381	-0.709	-2.594
	Max	0.006	0.144	0.584	0.317	0.440	0.006	0.144	0.584	0.317	0.440	0.010	0.144	0.613	0.341	0.440
	HE	0.002	0.002	-0.003	0.000	0.009	0.002	0.002	-0.004	0.000	0.008	0.001	0.003	-0.003	0.000	0.004
		BTC/GOLD														
DCC	Mean	-0.142	0.186	0.059	0.038	0.011	-0.142	0.189	0.060	0.042	0.014	-0.154	0.195	0.060	0.046	0.019
	Min	-0.446	-0.326	-1.157	-1.041	-12.900	-0.446	-0.314	-1.157	-1.041	-13.088	-0.460	-0.314	-1.157	-0.838	-12.900
	Max	0.063	3.486	2.769	3.542	3.987	0.063	3.805	2.894	3.963	5.164	0.063	5.400	3.039	3.982	5.476
	HE	0.008	-0.047	-0.020	-0.024	-0.017	0.009	-0.052	-0.021	-0.027	-0.023	0.009	-0.081	-0.022	-0.027	-0.025
ADCC	Mean	-0.130	0.208	0.094	0.071	0.045	-0.129	0.210	0.096	0.073	0.044	-0.138	0.211	0.096	0.072	0.051
	Min	-0.445	-0.285	-1.125	-0.999	-13.055	-0.445	-0.270	-1.125	-0.999	-13.564	-0.451	-0.270	-1.125	-1.011	-13.055
	Max	0.091	3.978	3.271	5.099	6.433	0.088	4.382	3.371	5.512	7.161	0.088	4.986	3.485	5.452	6.033
	HE	0.007	-0.057	-0.024	-0.040	-0.031	0.008	-0.063	-0.026	-0.044	-0.037	0.008	-0.074	-0.026	-0.045	-0.029
		BTC/OIL														
DCC	Mean	0.145	0.167	0.124	0.069	0.077	0.143	0.167	0.124	0.068	0.077	0.137	0.168	0.124	0.069	0.083
	Min	0.047	-0.045	-0.554	-0.526	-1.562	0.045	-0.045	-0.554	-0.526	-1.562	0.045	-0.045	-0.554	-0.574	-0.706
	Max	0.277	0.638	1.268	0.578	1.133	0.277	0.638	1.268	0.578	1.133	0.277	0.676	1.263	0.653	1.225
	HE	-0.005	-0.015	-0.006	-0.004	-0.006	-0.005	-0.016	-0.005	-0.003	-0.007	-0.005	-0.022	-0.006	-0.004	-0.008
ADCC	Mean	0.158	0.180	0.139	0.079	0.098	0.157	0.180	0.139	0.078	0.097	0.151	0.180	0.138	0.078	0.101
	Min	0.055	-0.030	-0.564	-0.543	-1.260	0.053	-0.030	-0.564	-0.543	-1.260	0.053	-0.030	-0.564	-0.607	-0.538
	Max	0.304	0.856	1.390	0.602	1.107	0.304	0.856	1.390	0.602	1.107	0.297	0.856	1.403	0.792	1.160
	HE	-0.006	-0.013	-0.005	-0.004	-0.005	-0.005	-0.015	-0.006	-0.004	-0.006	-0.005	-0.017	-0.006	-0.004	-0.007

Notes: This table presents the mean, minimum, and maximum values of the hedge ratios and hedge effectiveness values under the model refit (re: the number of periods to re-estimate the model)) in 10, 20, and 50 days, respectively. For each model refit, hedge ratios and hedge effectiveness were calculated by five different forecast lengths (fl) of 100, 500, 1,000, 1,500, and 2,000.

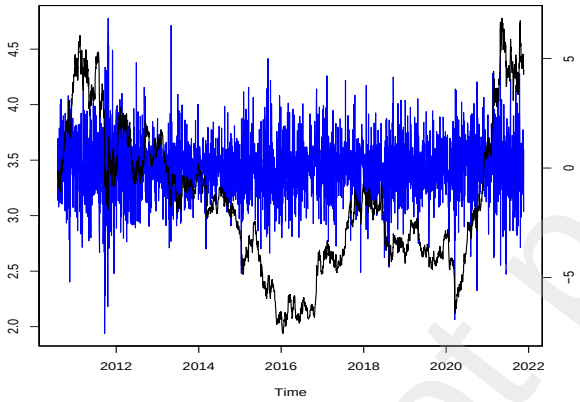
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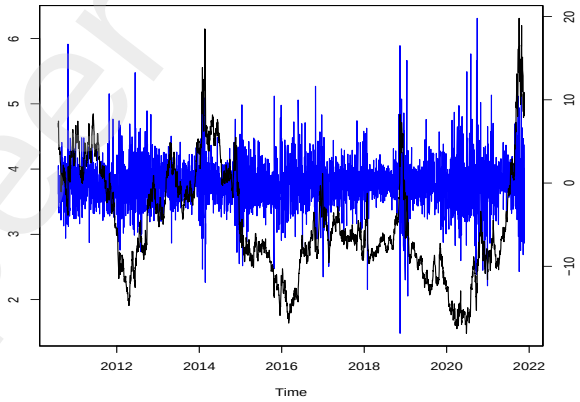
Figure 1: Price and Returns



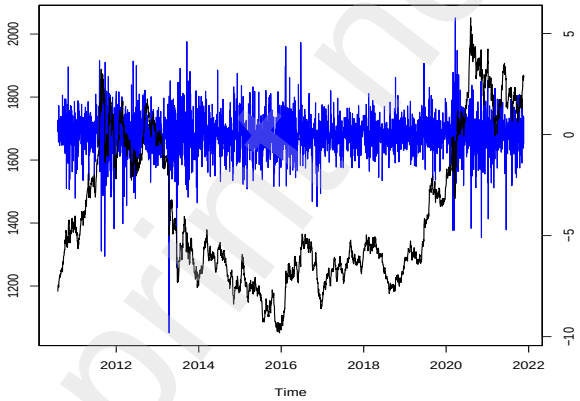
(1) BTC



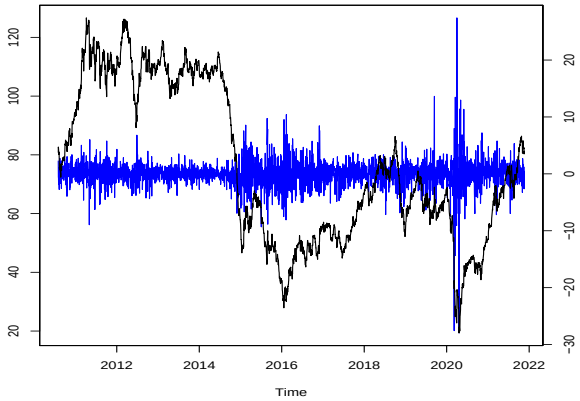
(2) Copper



(3) Gas



(4) Gold



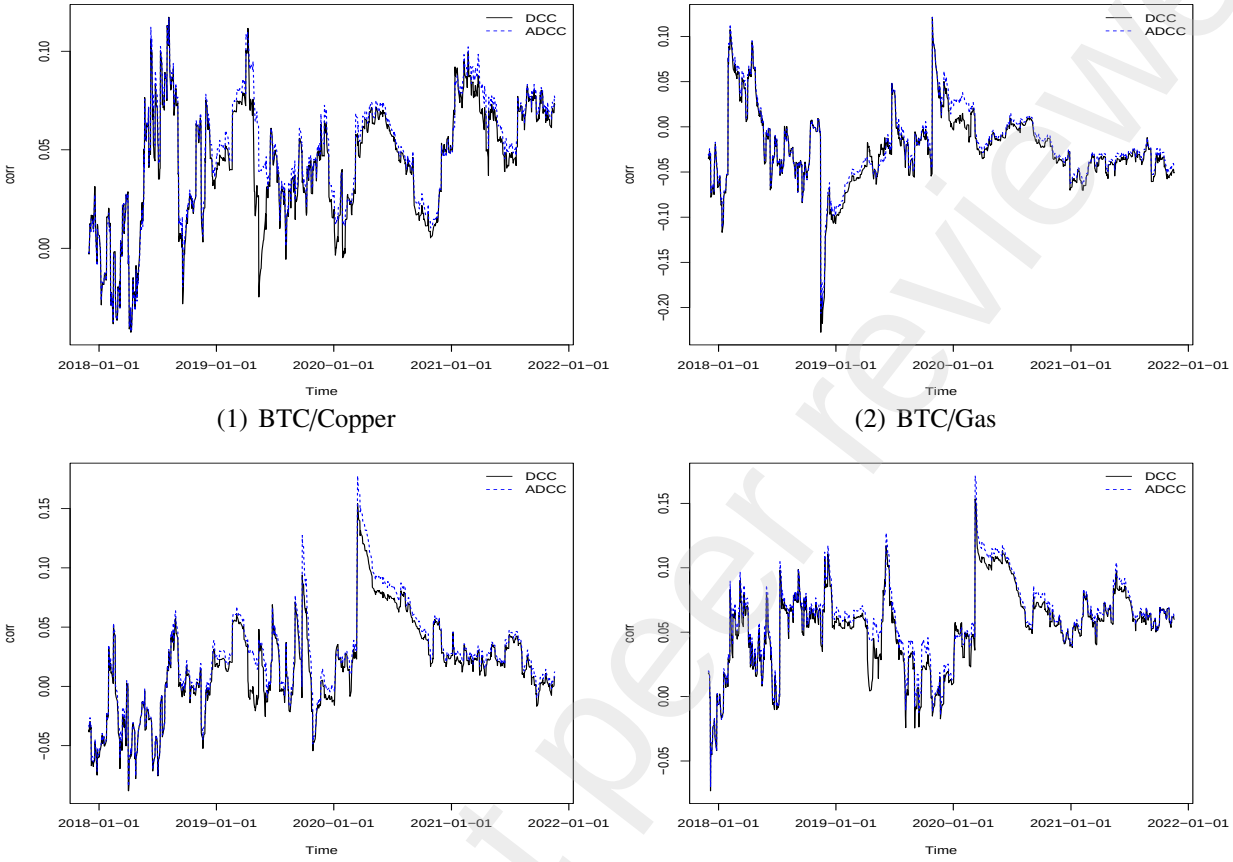
(5) Oil

Notes: The left hand side of vertical axes indicates the price, right hand side of vertical axes is log-price return, respectively.

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Figure 2: Dynamic Conditional Correlations



(1) BTC/Copper (2) BTC/Gas

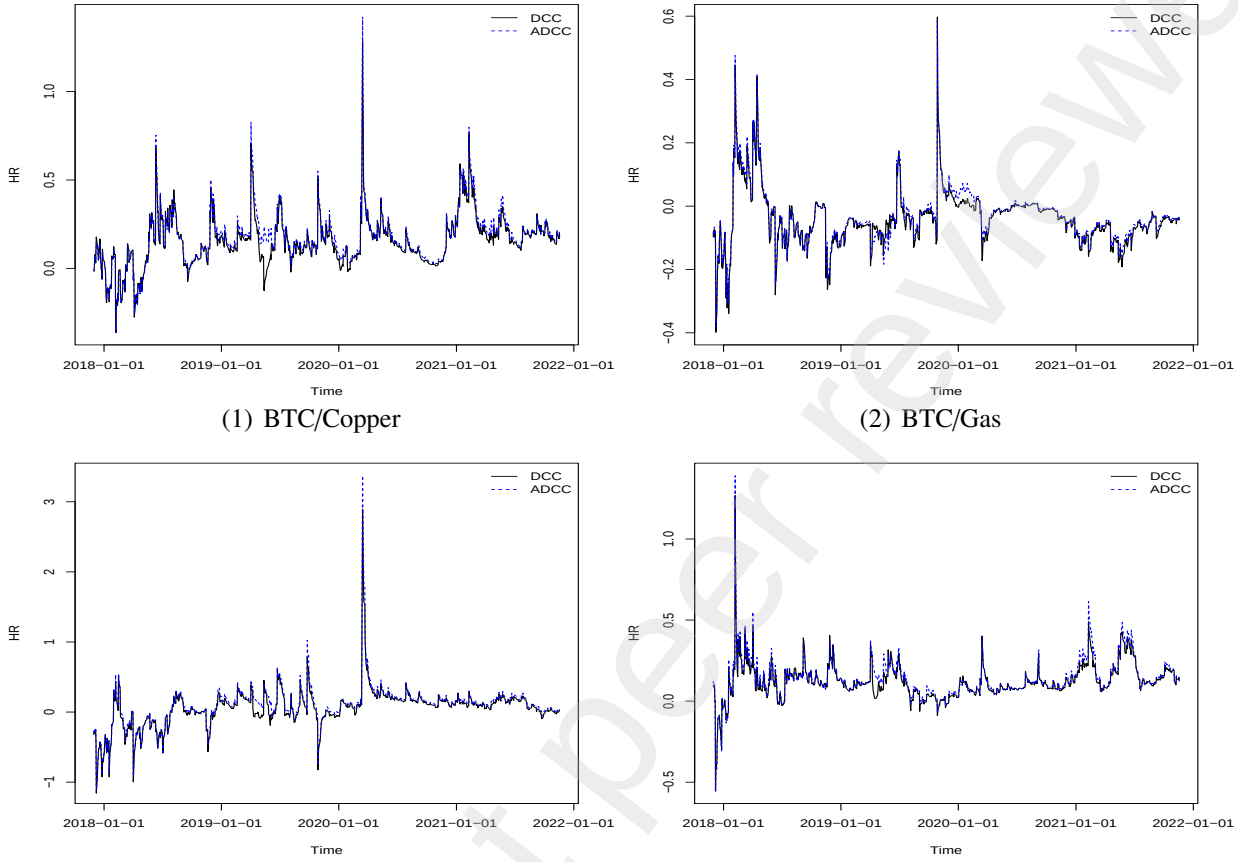
(3) BTC/Gold (4) BTC/Oil

Notes: The solid and dotted lines denote the dynamic conditional correlations from the DCC and ADCC models, respectively.

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Figure 3: Estimated Optimal Conditional Hedge Ratios



(1) BTC/Copper (2) BTC/Gas

(3) BTC/Gold (4) BTC/Oil

Notes: The solid and dotted lines denote the estimated optimal conditional hedge ratios from the DCC and ADCC models, respectively.

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